

The Perpetual That Rarely Pays: Funding Deadbands and Basis Discipline in the First CFTC-Regulated Perpetual Futures Market

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This version: 11 July 2026. First look and pre-registered analysis plan.

Abstract

On 29 May 2026 the CFTC approved Kalshi to list the first regulated perpetual futures contract in the United States. I provide, to my knowledge, the first empirical characterization of this market. Using the complete publicly available funding history through 11 July 2026 (1,226 eight-hour funding observations across 13 contracts, from 3 June 2026) and a purpose-built order book collector, I document three facts. First, the funding rate is exactly zero in 73.0% of funding periods. The mechanism explains how: the disclosed funding rule is a pure premium TWAP against the CF Benchmarks reference index with an explicit 0.01% zero threshold and no interest rate component. Second, the zero prints are not a mechanical artifact of the rounding rule. I apply Kalshi's rule counterfactually to Bybit's premium index over the identical window: the counterfactual zero share is 0.0%, because Bybit's BTC perpetual traded at a persistent discount to its index (mean premium of -4.7 basis points, never positive across five weeks) while Kalshi's perpetual centered on its externally administered reference index within one basis point in nearly two thirds of windows. Third, offshore realized funding appears compressed (mean absolute rate of 0.4bp on BTC) only because the premium and the clamped interest term nearly cancel; I validate this decomposition by reconstructing Bybit's realized funding from its public premium index to within 0.1bp. Two hypotheses can explain the onshore pattern: superior index tracking against a transparent, independently audited reference benchmark, or benchmark-anchored quoting by market makers in a young venue that may contribute little independent price information. These are observationally equivalent at funding frequency. I pre-register an information share analysis, to be executed on eight weeks of joint order book data, that separates them.

AI disclosure: the author used AI-assisted tools (Anthropic Claude) for code development, data analysis pipeline construction, and manuscript drafting and editing. The research questions, hypotheses, identification strategy, and interpretation of results are the author's; AI assistance did not generate the scientific claims. The author reviewed and verified all data, code, and text and takes full responsibility for the content.

Keywords: perpetual futures; funding rate; market design; deadband; CFTC; price discovery; cryptocurrency.
JEL: G13, G14, G18, G23.

1. Introduction

Perpetual futures are the dominant instrument in cryptocurrency trading. Industry estimates place global perpetual futures volume above ninety trillion dollars in 2025, up from roughly twenty-eight trillion in 2023. Until May 2026 no US-regulated venue offered the product. On 29 May 2026 the CFTC issued an order of approval to KalshiEX for a bitcoin perpetual contract, the first regulated perpetual futures contract in US history (CFTC, 2026). The exchange listed sixteen crypto perpetuals within six weeks;

press reports put cumulative perpetual volume above sixteen billion dollars by 9 July 2026 (Reuters, 2026).

A regulated perpetual is not an offshore perpetual with a different licence. The instrument's defining mechanism is the funding rate, and the design space is wide: the reference index, the periodicity, the clamps, and the premium measurement window all differ across venues. Kalshi anchors funding and settlement to CF Benchmarks indices, computes the rate as a pure time-weighted average of one-minute premiums, and snaps rates below 0.01% to zero. The dominant offshore design measures the premium against an internal mark price and adds a fixed interest baseline. These are different machines. This paper measures how differently they behave, in the first weeks in which the comparison has been possible.

I make three contributions. First, I document the funding behavior of the new market from its complete publicly available history: 73.0% of all funding observations are exactly zero, a frequency sharply different from the matched Bybit sample, in which no realized rate was exactly zero. The disclosed mechanism accounts for the possibility of a zero print, through its explicit deadband, but not for its frequency. Second, I show the deadband alone does not explain the frequency. Applying Kalshi's rule counterfactually to Bybit's own premium index over the identical window produces a zero share of 0.0%. The two venues occupy different equilibria: the offshore BTC perpetual traded at a persistent discount to its index throughout the sample, with a mean premium of -4.7 basis points and not a single eight-hour window of positive average premium, while the onshore perpetual centered on its index inside the one basis point band in 64.6% of windows. In the process I document a decomposition that is easy to misread: Bybit's realized funding looks near zero (mean absolute rate 0.38bp on BTC) only because a persistently negative premium and a positively clamped interest term nearly cancel. I validate this by reconstructing Bybit's realized funding from its public premium index under its stated formula to a mean absolute error of 0.1bp. Third, I provide, to my knowledge, the first description of the new venue's structure: contract design, listing waves, and market scale.

The central interpretive question is left deliberately open, because the funding data cannot answer it. Two hypotheses fit the onshore pattern. Under the first, which I call H-tight, arbitrage against a transparent, second-by-second, independently audited reference index is easy, so the basis rarely opens: regulation plus an externally administered benchmark delivers superior index tracking. Under the second, H-anchor, the venue is young and thin, market makers quote around the benchmark, and the premium is near zero by construction because the perpetual price is the index plus a spread; funding rarely activates because the venue may contribute little independent price information for it to discipline. The two are observationally equivalent at funding frequency. They separate sharply at higher frequency: H-anchor predicts the onshore venue contributes approximately nothing to price discovery. Section 7 pre-registers that test on eight weeks of joint order book data.

This version is a first look. The funding results use the complete funding history of the exchange. The order book collection began on 11 July 2026 and the basis and price discovery analyses are pre-registered rather than executed. I commit to reporting the pre-registered results without modification, and a finding of zero onshore price discovery contribution will be reported with the same prominence as its converse.

The paper sits at the junction of three literatures. The theory of perpetual futures runs from Shiller (1993) to the no-arbitrage treatments of He, Manela, Ross, and von Wachter (2024) and Akerer, Hugonnier, and Jermann (2025), in which the funding rate is the object that disciplines the perpetual toward its underlying; this paper studies a funding design in which that discipline is deliberately

switched off inside a stated band. Empirical work on crypto derivatives and cross-venue pricing documents persistent deviations and limits to arbitrage across venues (Makarov and Schoar, 2020) and price discovery leadership by offshore perpetuals (Alexander, Choi, Park, and Sohn, 2020); the pre-registered analysis in Section 7 asks where the first regulated onshore perpetual enters that ordering, using the information share machinery of Hasbrouck (1995) and Gonzalo and Granger (1995). The market design contribution is the documentation and decomposition of the deadband itself, which to my knowledge has no counterpart in the offshore designs studied to date.

2. Institutional background

2.1 The approval and the venue

The CFTC approved Kalshi's bitcoin perpetual on 29 May 2026 under Section 5c(c)(4) of the Commodity Exchange Act and Commission Regulation 40.3 (CFTC, 2026). On the same day, Commission staff issued interpretive and no-action relief addressing access to certain perpetual contracts treated as foreign futures under Commission Regulation 30.1, a separate regulatory route from Kalshi's directly listed DCM contract (CFTC staff letter, 2026). Kalshi's contracts trade directly on its designated contract market and clear through Kalshi Klear, a CFTC-regulated derivatives clearing organization. The accompanying policy statement indicated that case-by-case review under Regulation 40.3 is appropriate for perpetual contracts referencing asset classes not contemplated by the approval order. At the time of writing the exchange lists sixteen perpetual contracts: BCH, BTC, DOGE, DOT, ETH, HBAR, HYPE, LINK, LTC, NEAR, SHIB, SOL, SUI, XLM, XRP, and ZEC.

2.2 Contract design

Contracts are small and linear. The bitcoin perpetual represents 0.0001 BTC per contract, roughly six dollars of notional at sample-period prices, quoted in decimal dollars to four places, USD-margined, trading around the clock. Reference prices for funding and settlement come from CF Benchmarks real-time indices; for BTC this is the Bitcoin Real-Time Index (BRTI), which aggregates regulated spot venues and updates every second. Leverage is capped, varies by asset and position size, and is exposed per size tier through the API together with a liquidation mark price. Margin is isolated in the retail applications and portfolio-based via the API.

2.3 The funding mechanism

The funding rule is fully disclosed. Every eight hours the exchange computes a time-weighted average of one-minute candlestick premiums over the 480 candles in the funding period, where the premium is the gap between the perpetual price and the reference index. The resulting rate is clamped at plus or minus 2% per interval. Two features distinguish the design from the offshore standard. First, there is no interest rate component: the rate is a pure premium TWAP, whereas the dominant offshore formula adds a fixed 0.01% per eight hours interest baseline to the premium via a clamped adjustment term. Second, there is an explicit zero threshold: if the absolute rate is below 0.01%, it is set to zero and no payment is made. The deadband documented in Section 4 is therefore a stated rule, not an inference. A third disclosed feature matters for robustness: if the CF Benchmarks feed is unavailable, funding pauses until the feed is restored, so the mechanism has an explicit reference-feed failure mode. One measurement detail is not fully specified in the public documentation: the exchange describes one-minute candlestick premiums, while the CFTC order describes funding as based on the difference between the contract's mark price and the reference price; which exchange-side price series enters the candles (trade, mark, or midpoint) is therefore verified empirically as part of the registered test in Section 4.4.

Funding accrues at 12:00 AM, 8:00 AM, and 4:00 PM Eastern Time, which is 04:00, 12:00, and 20:00 UTC during daylight saving and shifts by one hour each November. The major offshore venues pay at 00:00, 08:00, and 16:00 UTC year-round. The four-hour stagger, and the DST shift itself, provide identification for any funding-time effects on prices or flow. Settlement cycles run at 12:00 PM and 4:00 PM ET through Kalshi Klear.

Table 1. Funding mechanism and contract design: Kalshi vs the offshore standard.

Feature	Kalshi	Bybit
Reference price for funding	CF Benchmarks index (externally administered, independently audited, 1s)	premium index from impact bid/ask vs underlying index
Premium measurement	TWAP of 480 x 1-min premiums	average of 1-min premium index over 8h
Interest component	none	0.01%/8h via clamped term
Zero threshold	0.01%/8h (stated)	none
Funding cap	$\pm 2\%/8h$	instrument-specific clamp (instruments endpoint)
Funding times (UTC)	04/12/20 (DST: 05/13/21)	00/08/16
Contract size (BTC)	0.0001 BTC/contract	qty in BTC (min 0.001)
Clearing	Kalshi Klear (CFTC DCO)	internal
Public trades feed	no	yes

Bybit realized funding follows $F = P + \text{clamp}(I - P, \pm 0.05\%)$ with $I = 0.01\%$ per 8h (Bybit funding methodology documentation); the formula is verified against realized prints in Appendix C to 0.1bp mean absolute error, which is the operative check for this paper. Kalshi rows from exchange documentation accessed 11 July 2026 (archived); Bybit rows from Bybit v5 documentation. Bybit distinguishes the underlying index, impact bid and ask prices, the premium index, and the mark price; the premium index is the funding input.

3. Data

3.1 Kalshi

All Kalshi data come from public, unauthenticated REST endpoints under the exchange's margin namespace. I collect three objects. First, the complete publicly available funding history through 11 July 2026: 1,226 funding observations across 13 contracts, from the first funding event at 20:00 UTC on 3 June 2026 through 04:00 UTC on 11 July 2026. The exchange serves the full history in one response per ticker. The earliest observation for every launch contract is 3 June even though the bitcoin contract was approved 29 May, which indicates funding activated after a short launch window; I treat 3 June as the start of the funding sample. Three of the sixteen listed contracts (DOT, HBAR, XLM) have no funding observations at the sample freeze, consistent with listing within days of the freeze date. Second, full two-sided order books, polled every two seconds per contract from 11 July 2026. Third, exchange snapshots of price, bid, ask, open interest, and notional open interest, polled every five seconds. The venue exposes no public trades feed; per-interval traded volume is recoverable by differencing the cumulative volume fields in consecutive snapshots (Appendix B).

3.2 Bybit

Offshore comparators come from Bybit's public v5 API over the identical window: realized eight-hour funding for the thirteen matched linear perpetuals, and one-minute premium index klines for BTCUSDT and ETHUSDT (roughly 54,000 candles each), which are the direct input to Bybit's funding formula. Appendix C validates that the premium series reconstructs Bybit's realized funding under its stated formula to a mean absolute error of 0.10bp (BTC) and 0.13bp (ETH), with correlation 0.91 and 0.92 between realized rates and window-average premiums. The validation matters: it certifies that the counterfactual in Section 4 operates on the same object Bybit's own funding engine consumes.

Table 2. Funding summary, 3 June to 11 July 2026.

Series	N	Zero share	Mean rate , nonzero (bp)	First obs
Kalshi KXBTCPERP	113	0.646	1.65	2026-06-03
Kalshi KXETHPERP	113	0.770	1.84	2026-06-03
Kalshi KXSOLPERP	113	0.929	2.63	2026-06-03
Kalshi KXXRPPERP	113	0.566	2.14	2026-06-03
Kalshi KXHYPEPERP	98	0.643	2.06	2026-06-08
Kalshi KXBCHPERP	95	0.621	2.70	2026-06-09
Kalshi KXDOGEPERP	95	0.905	1.56	2026-06-09
Kalshi KXKSHIBPERP	95	0.547	1.76	2026-06-09
Kalshi KXLINKPERP	97	0.990	1.45	2026-06-09
Kalshi KXLTCPERP	95	0.674	1.35	2026-06-09
Kalshi KXSUIPERP	95	0.653	1.51	2026-06-09
Kalshi KXNEARPERP	52	0.808	1.42	2026-06-24
Kalshi KXZECPERP	52	0.808	1.67	2026-06-24
Kalshi pooled	1,226	0.730	1.88	2026-06-03
Bybit BTCUSDT	113	0.000	0.38	2026-06-03
Bybit ETHUSDT	113	0.000	0.39	2026-06-03
Bybit, 13 matched symbols pooled	1,469	0.000	0.38–2.34 by symbol	2026-06-03
Counterfactual: Kalshi rule on Bybit BTC premium	113	0.000	4.72	2026-06-03
Counterfactual: Kalshi rule on Bybit ETH premium	113	0.000	4.88	2026-06-03

Kalshi rates from the exchange funding history endpoint. Bybit realized rates from the public funding history endpoint. Counterfactual rows apply Kalshi's stated rule (8h TWAP of 1-min premiums on Kalshi's ET funding grid, $\pm 2\%$ clamp, 0.01% zero snap) to Bybit's premium index klines. Rates per 8h; 1bp = 0.01%.

4. The funding deadband

4.1 The fact

Of 1,226 funding observations across thirteen contracts, 73.0% are exactly zero. The stated mechanism explains how a zero print arises: the eight-hour TWAP premium against the reference index fell inside the disclosed 0.01% deadband and was snapped to zero. Conditional on activation, the mean absolute rate is 1.88bp, barely above the 1bp threshold, so the premium distribution is massed near zero and the deadband clips its shoulder. The clipped shoulder is one-sided: the pooled distribution of nonzero prints concentrates just below the band, between -1 and -3bp, with only a thin positive tail (Figure 2): 253 of the 331 nonzero prints (76.4%) are negative. The deadband censors a distribution centered slightly below zero. The cross-section is wide: zero shares range from 54.7% (SHIB) and 56.6% (XRP) to 92.9% (SOL) and 99.0% (LINK), which Section 6 relates to activity. Because funding observations are serially

dependent within contracts, block-bootstrap intervals for the zero shares are registered alongside the Section 7 analysis rather than reported here.

The offshore comparison cannot be made on realized rates. Bybit's realized funding over the identical window never prints zero on any of the thirteen matched symbols, but this is uninformative about the basis, because the offshore formula embeds a fixed interest baseline and has no rounding rule. The comparison has to be made at the level of the premium itself.

4.2 Counterfactual rule swap

I apply Kalshi's rule to Bybit's own premium index: the average of one-minute premium index closes over each eight-hour window on Kalshi's ET funding grid, clamped at $\pm 2\%$, snapped to zero below 0.01%. If the offshore counterfactual zero share were close to 73%, the zero prints would be a property of the rounding rule applied to any tight market. The result is the opposite. The counterfactual zero share is 0.0% for both BTC and ETH: in 113 windows over five weeks, Bybit's average premium never once fell inside the one basis point band. The mean absolute counterfactual rate is 4.72bp (BTC) and 4.88bp (ETH), roughly five times the width of the band. The deadband alone therefore does not explain the 73% zero share; the share reflects the interaction between the rule and Kalshi's unusually concentrated near-zero premium distribution. The result is not specific to the Eastern Time grid: on Bybit's native 00/08/16 UTC funding grid, the window-average premium falls inside the one basis point band in 0 of 115 windows for both BTC and ETH (Appendix C sample), so the counterfactual zero share is 0.0% on either venue's clock. A rolling-window version is registered with Section 7 as a further check.

4.3 A sign asymmetry, and a decomposition that is easy to misread

The premium series shows why. Bybit's BTC premium is strictly negative over the entire sample: the mean one-minute premium is -4.7bp, and the maximum is -0.07bp, so the perpetual traded below its index in effectively every minute of five weeks. The ETH premium is negative on average (-4.9bp) with brief positive excursions to +11.8bp. One possible interpretation is that bearish positioning and venue-specific carry demand sustained the discount during the broad bitcoin drawdown of the first half of 2026; the funding data alone do not identify the channel. Realized Bybit funding nevertheless averages only 0.38bp in absolute value, because the formula's clamped interest term contributes up to +5bp against a premium near -4.7bp: two large terms nearly cancel. Small realized offshore funding is not evidence of a tight basis. Appendix C verifies the decomposition by reconstructing realized funding from the premium series to 0.10bp mean absolute error.

The onshore market occupies a different equilibrium. Kalshi's perpetual shows no persistent discount: it centers on its external reference index, inside the one basis point band in 64.6% of BTC windows, with excursions in both directions. Conditional on escaping the band, the pooled prints lean negative (76.4% of nonzero events), the same direction as the offshore discount but roughly one fifth its magnitude, so the onshore market shares the sign of the drawdown-regime carry while sitting almost entirely inside the band. The two largest BTC excursions in the sample are +9.85bp at the 04:00 UTC funding on 5 June, in the first week of funding, and -7.94bp at 12:00 UTC on 18 June; the next largest prints are an order of magnitude smaller (+2.46bp on 3 July, +2.15bp on 26 June, +2.04bp on 18 June), and from late June onward the nonzero prints are persistently small and positive, between 1 and 2.5bp. Same asset, same sample period, markedly different premium distributions: the offshore perpetual sustains a persistent discount against its premium index; the onshore perpetual sits on its externally administered reference index.

4.4 Verification of the stated rule (registered)

The remaining test is verification: reconstructing Kalshi's TWAP premium from collected mid-quotes and the reference index and confirming that realized prints equal the stated rule, clamp then snap. This requires the order book sample and an index leg, so I register it here and execute it with the Section 7 analysis. Discrepancies, if any, will be reported.

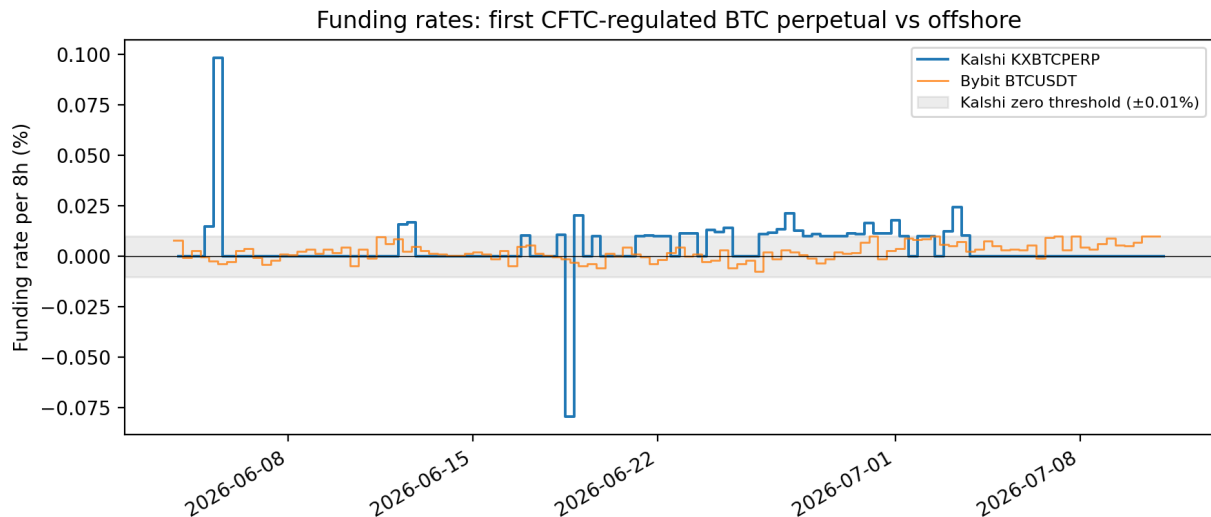


Figure 1. Realized eight-hour funding rates, 3 June to 11 July 2026: Kalshi KXBTCERP (blue) vs Bybit BTCUSDT (orange). Grey band: Kalshi's $\pm 0.01\%$ zero threshold. The Kalshi rate sits at exactly zero in 64.6% of windows; the Bybit rate never prints zero but is compressed by the near-cancellation of a persistently negative premium and the clamped interest term (Section 4.3).

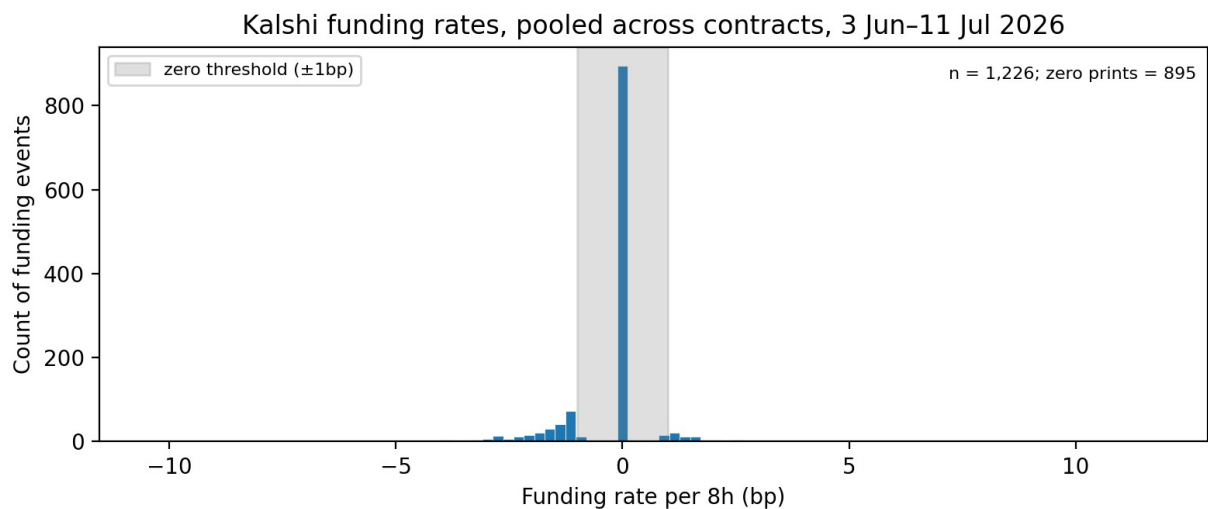


Figure 2. Kalshi funding rates pooled across contracts, 3 June to 11 July 2026 ($n = 1,226$; 895 zero prints). Grey band: the $\pm 1bp$ zero threshold. The nonzero mass concentrates on the negative shoulder just outside the band; the positive tail is thin. The deadband censors a distribution centered slightly below zero.

5. Two hypotheses, one test

Why does the onshore perpetual sit on its index? Two mechanisms fit everything in Section 4. Under H-tight, the reference index is transparent, second-by-second, and externally audited, so arbitrage against it is cheap and fast; deviations are corrected before they accumulate into an eight-hour TWAP outside the band. Under H-anchor, the venue is young and thin, its market makers quote around the benchmark, and the perpetual price is the index plus a spread by construction: the premium is near zero because little else

enters it, and funding rarely activates because the venue may contribute little independent price information for it to discipline. The offshore market illustrates a venue in which the perpetual sustains a material and persistent deviation from its reference index; whether that deviation reflects independent information or other channels (inventory, leverage demand, collateral constraints, segmentation) is itself a question for the information share analysis.

The hypotheses are observationally equivalent in funding data. They separate at higher frequency. H-anchor predicts the onshore venue contributes approximately nothing to price discovery: its mid-quote should follow the index and the offshore complex with a lag and an information share near zero. H-tight is consistent with a small but nonzero contribution. Three auxiliary predictions are testable in the first-look sample as it accumulates and are registered alongside the main test: under H-anchor, the onshore mid should never lead the index at any horizon; deadband exits should be driven by index volatility rather than onshore order flow; and the cross-sectional zero share should fall as venue activity rises, weakening the anchor where trading thickens. The cross-section in Table 2 is suggestive on the last point but confounded by listing age, so I defer it to the executed version.

Either resolution is informative for design. If H-tight survives, a deadband over an independently audited external benchmark avoids funding churn without losing price fidelity, and the mechanism is attractive beyond this venue. If H-anchor survives, the 73% is the signature of a satellite market, and the interesting question becomes whether the anchor loosens as the venue matures, which the continuing collection will show.

6. Listing waves and market scale

The venue listed in waves. Funding activated for the launch cohort (BTC, ETH, SOL, XRP) on 3 June, five days after approval; HYPE followed on 8 June; a six-contract wave (BCH, DOGE, SHIB, LINK, LTC, SUI) on 9 June; NEAR and ZEC on 24 June; and DOT, HBAR, and XLM listed close enough to the freeze to have no completed funding period. The market is small in absolute terms: at the 11 July 2026 snapshot, BTC open interest is 693,877 contracts, or 4.45 million dollars notional, with a quoted spread of roughly 2bp at the touch. Full microstructure descriptives, including quoted and volume-differenced effective spreads, depth at 10, 25, and 50 basis points normalized to dollar notional, and open interest growth across the sixteen contracts, require the accumulating order book sample and are registered with the Section 7 analysis; the cross-section of zero shares in Table 2 will be related to activity there, in a contract-by-funding-window panel with listing-age controls, and presented as exploratory given the small cross-section.

7. Pre-registered analysis: price discovery

I pre-register the following and will execute it without modification once eight weeks of joint order book data have accumulated, on or after 12 September 2026. Question: does the regulated onshore perpetual contribute to price discovery, or is it a satellite of the offshore complex and the spot index, as H-anchor predicts? Design: Hasbrouck information shares and Gonzalo-Granger component shares on one-second mid-quotes for BTC and ETH across Kalshi, Bybit, and the constituent regulated spot venues that the reference index aggregates, used individually rather than as a single index leg, because a calculated benchmark is an aggregation of constituent prices and not an independent trading venue; where the benchmark itself is used it is labelled as a reference process. Information shares are reported as Hasbrouck upper and lower bounds across covariance factorization orderings, estimated daily, with block-bootstrap confidence bands. Robustness: 5-second and 10-second sampling and event-time

estimation alongside the one-second baseline; lead-lag confirmation by the lagged Hayashi-Yoshida cross-covariance estimator of Hoffmann, Rosenbaum, and Yoshida (2013). Auxiliary tests: onshore-lead incidence at 1s to 60s horizons; deadband-exit attribution to index volatility versus onshore order flow (volume-differenced); cross-sectional zero share on activity with listing-age controls. Commitments: the freeze date, venue set, sampling frequency, and both estimators are fixed now; deviations will be reported as deviations. A finding of zero onshore contribution is a result, not a failure, and will be reported with the same prominence as its converse.

8. Conclusion

The first CFTC-regulated perpetual futures market is a different machine from its offshore ancestors, and five weeks of its publicly available funding history already show the difference. Its funding rate is exactly zero in 73.0% of recorded funding intervals, under a disclosed rule with a pure premium TWAP, an explicit one basis point deadband, and no interest component. The deadband alone does not explain the zeros: the same rule applied to a major offshore perpetual's own premium index never produces a zero over the identical window, because the offshore contract carried a persistent five basis point discount to its premium index while the onshore contract sat on its externally administered reference index. Whether that reflects superior index tracking under regulation or benchmark-anchored quoting in a venue that contributes little independent price information is the question the funding data cannot answer, and the pre-registered information share analysis will.

Appendix A. Data collection

Funding history, order books, and market state are collected from the production margin namespace at `external-api.kalshi.com/trade-api/v2/margin/`, which serves market data without authentication; the API documentation described the production margin API as unreleased at the access date, and access dates are logged. Books are polled at two-second cadence per contract to hourly-rotated compressed JSONL; market state (price, bid, ask, open interest, notional, full raw JSON) at five seconds to SQLite; the funding estimate each minute and the funding history hourly. All receipt timestamps come from one NTP-disciplined host. Bybit data come from the public v5 endpoints `funding/history` and `premium-index-price-kline`. Code sufficient to reproduce all tables and figures is available from the author; the full host, clock discipline, and replication checklist for the order book sample accompany the executed Section 7 version.

Appendix B. Volume recovery by snapshot differencing

The venue exposes no public trades feed. Per-interval traded volume is recovered by differencing the cumulative volume fields in consecutive market state snapshots at five-second cadence. Validation: consistency of recovered daily volume with the exchange's 24-hour volume fields, and consistency of open interest changes across funding boundaries. Failure modes and their handling: missed snapshots (interval flagged), field resets (detected as negative differences). The validation is executed with the Section 7 analysis, since it requires the accumulated collector sample.

Appendix C. Validation of the Bybit premium series

The Section 4 counterfactual is valid only if the premium index klines are the object Bybit's funding engine consumes. For each realized funding event I reconstruct the rate from the one-minute premium series under the stated formula $F = P + \text{clamp}(I - P, \pm 0.05\%)$ with $I = 0.01\%$ per eight hours, where P is

the window-average premium. Table C1 reports the fit against two alternatives. The reconstruction runs on Bybit's native 00/08/16 UTC funding grid from 00:00 UTC on 3 June, which yields 115 evaluated windows; 113 of these lie within the common comparison sample of Table 2, which begins at Kalshi's first funding event (20:00 UTC on 3 June). Each eight-hour window contains up to 480 expected one-minute candles; windows with fewer than 240 observed candles are excluded (one window at the sample start), duplicate timestamps are dropped, and no interpolation is applied. The stated formula reproduces realized funding to a mean absolute error of 0.10bp (BTC) and 0.13bp (ETH); the alternatives miss by roughly 4-5bp. Correlation between realized rates and window-average premiums is 0.911 and 0.922. The premium series is validated, and the decomposition in Section 4.3, near-cancellation of a negative premium and the positively clamped interest term, follows directly.

Table C1. Reconstruction of Bybit realized funding from the public premium index, Bybit funding grid, 3 June to 11 July 2026 (115 windows; 113 within the common Table 2 sample).

Symbol	Windows	MAE, stated formula (bp)	MAE, pure premium (bp)	MAE, premium + 0.01% (bp)	corr(F, mean premium)
BTCUSDT	115	0.098	4.960	3.960	0.911
ETHUSDT	115	0.134	4.995	3.995	0.922

Premium series: 1-min premium-index-price klines (about 54,000 per symbol). BTC premium over the window: mean - 4.72bp, range [-12.78, -0.07]bp. ETH: mean -4.88bp, range [-35.70, +11.84]bp.

Disclosure

The author used AI-assisted tools (Anthropic Claude) for code development, data analysis pipeline construction, and manuscript drafting and editing. The research questions, hypotheses, identification strategy, and interpretation of results are the author's; AI assistance did not generate the scientific claims. The author reviewed and verified all data, code, and text and takes full responsibility for the content. No funding was received for this work. The author declares no competing interests. Data and code sufficient to reproduce all tables and figures are available from the author.

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